

Application To Make Gas Filling Station Easy Using CRM (Developer)

ABSTRACT:

Gas filling stations often face challenges in maintaining accurate records, ensuring smooth operations, and delivering quick service to customers. To address these issues, this project leverages Salesforce CRM to design a customized application that centralizes and streamlines the entire process.

The solution introduces dedicated objects such as Buyer, Supplier, Fuel Details, and Gas Station to store and manage critical information in one place. With tailored page layouts capturing personal details, vehicle information, and billing data, the application reduces manual errors and enhances transparency.

User roles and permission sets were also implemented to define responsibilities clearly and maintain data security. Along with these, features like automated workflows, validation rules, and access controls strengthen reliability and efficiency.

Overall, the project demonstrates how Salesforce CRM can be applied beyond traditional domains by creating a practical solution for gas stations, improving both operational management and customer satisfaction.

OBJECTIVE:

The primary purpose of this project is to design a Salesforce-based CRM solution that transforms the way gas filling stations handle their operations. Instead of relying on traditional manual methods, the application focuses on digitalization, automation, and secure data handling.

The project is guided by the following objectives:

- **Digital Transformation:** Replace paper-based records with a cloud-driven CRM that stores and manages complete information on Buyers, Fuel Details, Gas Stations, and Suppliers in real time.
- **Customer-Centric Service:** Deliver a smoother experience to customers by maintaining structured records of buyer details, vehicle information, fuel purchases, and payment history.

- **Operational Transparency:** Provide managers and staff with clear visibility of supplier contributions, fuel stock levels at different gas stations, and daily transactions.
- **Error Reduction:** Minimize mistakes in billing, fuel entry, and supplier data by using automated flows, validation rules, and formula fields.
- **Secure Collaboration:** Enable different roles—Managers, Sales Executives, and Salespersons—to work together efficiently with role-specific permissions while protecting sensitive data.
- **Future Readiness:** Lay the groundwork for advanced features such as dashboards, AI-powered insights, and mobile accessibility to meet the growing needs of gas station management.

TECHNOLOGY DESCRIPTION:

This project has been developed on **Salesforce CRM**, a cloud-based solution designed to enhance customer relationship management and optimize business processes. The platform provides a wide range of features for building custom applications, defining user hierarchies, safeguarding data, and generating actionable reports.

The key components applied in the Gas Station CRM include:

Custom Objects:

Business-specific objects such as Buyer, Fuel Details, Gas Station, and Supplier were created to store and manage essential data. These objects serve as structured repositories, ensuring organized and efficient record handling.

Tabs:

Dedicated tabs were configured for each custom object, enabling quick navigation and direct access to critical information within the Lightning App.

Lightning App:

A specialized application named Gas Station was designed, integrating related objects, tabs, and utilities into one streamlined interface for users to manage daily operations effectively.

Page Layouts:

Fields were grouped into logical sections like personal details, vehicle data, fuel transactions, and receipts. This improved data clarity, minimized errors, and simplified data entry.

Profiles and Roles:

Access control was implemented through tailored profiles and hierarchical roles. Distinct roles such as Manager, Sales Executive, and Salesperson were defined to distribute responsibilities.

Permission Sets and Organization-Wide Defaults (OWD):

Permission sets provided additional access flexibility beyond profiles, while OWD established baseline record-level visibility, strengthening security across the system.

Reports and Dashboards:

Reporting tools and dashboards were configured to track performance indicators such as fuel usage, supplier contributions, sales activity, and customer interactions, offering real-time business insights.

Flows and Automation:

Automated flows handled repetitive tasks like sending fuel purchase receipts and updating transaction details, reducing dependency on manual processes.

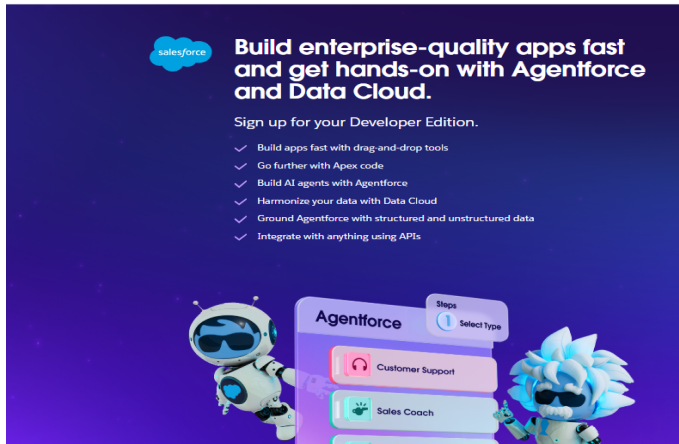
Apex Triggers:

Custom triggers were introduced to apply advanced business logic, including validation of fuel prices and restrictions on improper record deletion. Together, these Salesforce features create a robust and scalable CRM solution for managing gas station operations, ensuring smooth coordination among buyers, suppliers, and employees while maintaining efficiency and security.

Project Implementation Steps :

Step 1 — Setting up a Salesforce Developer Org :

1. Open the Developer sign-up page and go to the Salesforce Developer sign-up and fill in the form with your details (name, email, company, country, and a username in email format).
2. Choose and confirm a password that meets Salesforce's password rules, then submit the form.



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First name: doma Last name: rohith

Job title: developer Work email: 22kd1a0229@lendl

Company: lendl Country/Region: India

☒ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement. I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features, and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

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☒ I'm not a robot

- Verify your email – check your inbox (and spam folder) for Salesforce's verification message and click the link to confirm your account.

Welcome to **Salesforce**: Reset your password External Inbox x

support@salesforce.com <support@salesforce.com>
to me ▾

Wed, Sep 3, 1:24 PM ☆ ↶ ⋮



Hi doma,

Thanks for signing up for a Developer Edition. Now you can start building on **Salesforce** for free and get hands-on with Agentforce and Data Cloud.

There's just one more step. Use the following link to reset the password for your Developer Edition. This link expires in 24 hours.

[Reset Password](#)

To easily log in later, save this URL:
<https://orgfarm-52980b5ce7-dev-ed.develop.my.salesforce.com>

Here's the username for your Developer Edition:
22kd1a0229739@agentforce.com

Your Developer Edition, now enabled with Agentforce and Data Cloud, remains active as long as you continue to use it. It expires after 45 days of non-usage.

Step 2 – Creating Custom Objects:

Procedure:

1. In Salesforce, go to Setup → Object Manager → Create → Custom Object.
2. Provide an Object Label and Object Name (the API name will auto-populate).

3. Enable Allow Reports and Track Field History if you need reporting and change tracking.
4. Save the object, then repeat these steps for each remaining entity.

Custom Objects Required

- **Buyer__c** – Stores customer details and fuel purchase records.
- **Fuel_Details__c** – Logs fuel transactions, linking buyers, suppliers, and gas stations.
- **Gas_Station__c** – Maintains station data, fuel availability, and usage statistics.
- **Supplier__c** – Tracks supplier information and total quantities supplied to stations.

The screenshot displays the Salesforce Setup interface. At the top, there is a search bar labeled 'Search Setup' and a navigation bar with 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' section is expanded, showing a list of objects. The 'Buyer' object is selected, and its configuration details are shown on the right. The configuration includes fields for 'API Name', 'Singular Label', 'Plural Label', 'Enable Reports', 'Track Activities', 'Track Field History', 'Deployment Status', and 'Help Settings'. The 'Buyer__c' API name is entered, and the 'Singular Label' is 'Buyer' and the 'Plural Label' is 'Buyers'. The 'Enable Reports' and 'Track Field History' checkboxes are checked. The 'Deployment Status' is 'Deployed'. The 'Help Settings' are set to 'Standard salesforce.com Help Window'. There are 'Edit' and 'Delete' buttons in the top right corner of the configuration area.

SETUP > OBJECT MANAGER

Buyer

[Edit](#) [Delete](#)

Details

Description

API Name
Buyer__c

Custom
✓

Singular Label
Buyer

Plural Label
Buyers

Enable Reports
✓

Track Activities

Track Field History
✓

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window



Q Search Setup



SetupHomeObject Manager

SETUP > OBJECT MANAGER

Fuel details

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Details

Description

API Name

Fuel_details__c

Custom

✓

Singular Label

Fuel details

Plural Label

Fuel details

Enable Reports

✓

Track Activities

Track Field History

✓

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

EditDelete



Q Search Setup



SetupHomeObject Manager

SETUP > OBJECT MANAGER

Gas Station

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

Details

Description

API Name

Gas_Station__c

Custom

✓

Singular Label

Gas Station

Plural Label

Gas Stations

Enable Reports

✓

Track Activities

Track Field History

✓

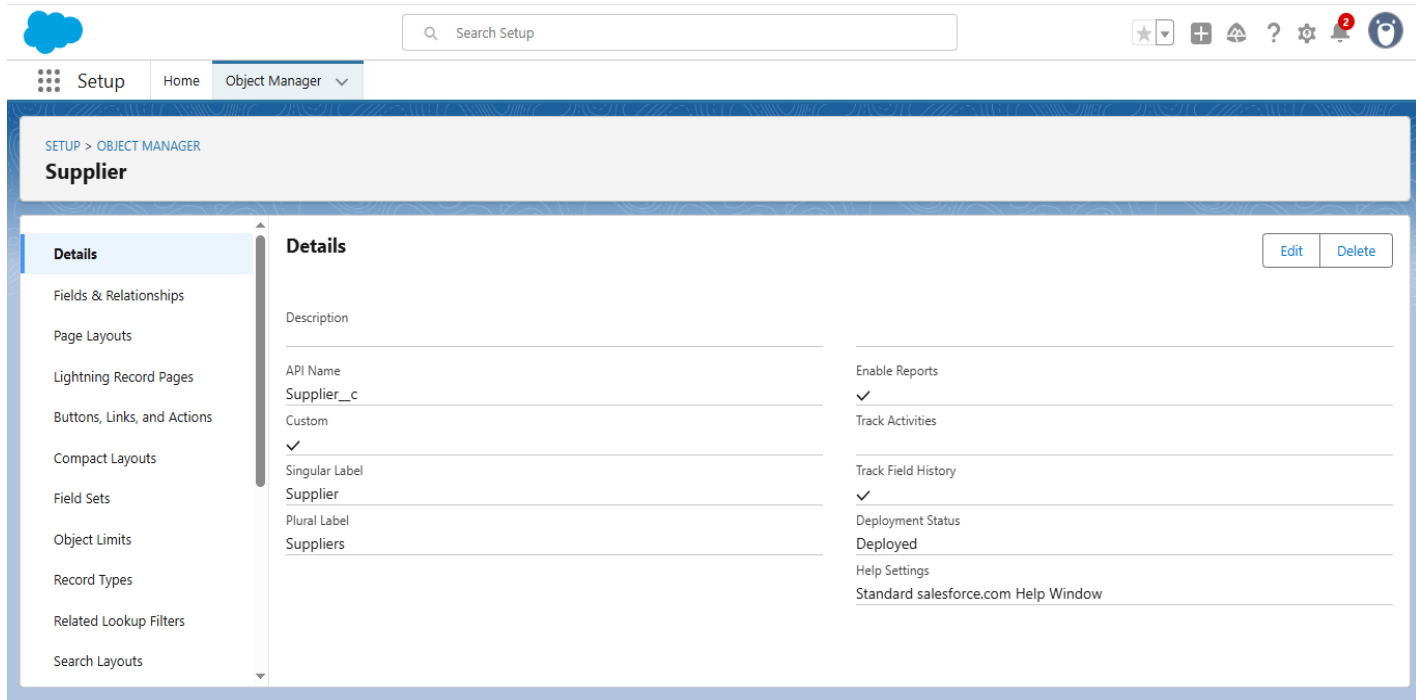
Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

EditDelete



Step 3 – Creating Custom Tabs:

Purpose :

To provide quick and intuitive navigation for users, making it easy to access and manage records for the custom objects within the Gas Station CRM.

Procedure:

1. In Salesforce, navigate to Setup → Tabs → New.
2. Under Custom Object Tabs, choose the custom object you want to create a tab for (e.g., Buyer, Fuel Details, Gas Station, or Supplier).
3. Select a Tab Style (icon and color) that represents the object clearly.
4. Choose the user profiles that will have access to this tab.
5. Click Next, review your settings, and then Save.
6. Repeat these steps for each remaining custom object so all four objects have their own dedicated tabs.

Tabs Created:

- **Buyer Tab** – For customer information and transactions.
- **Fuel Details Tab** – To view and manage fuel supply records.
- **Gas Station Tab** – For station-specific data and fuel availability.
- **Supplier Tab** – To monitor supplier details and fuel delivery information.

Setup Home Object Manager ▾

Q tabs

▾ User Interface

Rename **T**abs and Labels

Tabs

Didn't find what you're looking for?
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Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs [New](#) [What Is This?](#)

Action	Label	Tab Style	Description
Edit Del	Buyers	Bottle	
Edit Del	Fuel details	Factory	
Edit Del	Gas Stations	Globe	
Edit Del	Suppliers	Safe	

Web Tabs [New](#) [What Is This?](#)

No Web Tabs have been defined

Step 4 — Creating a Lightning App:

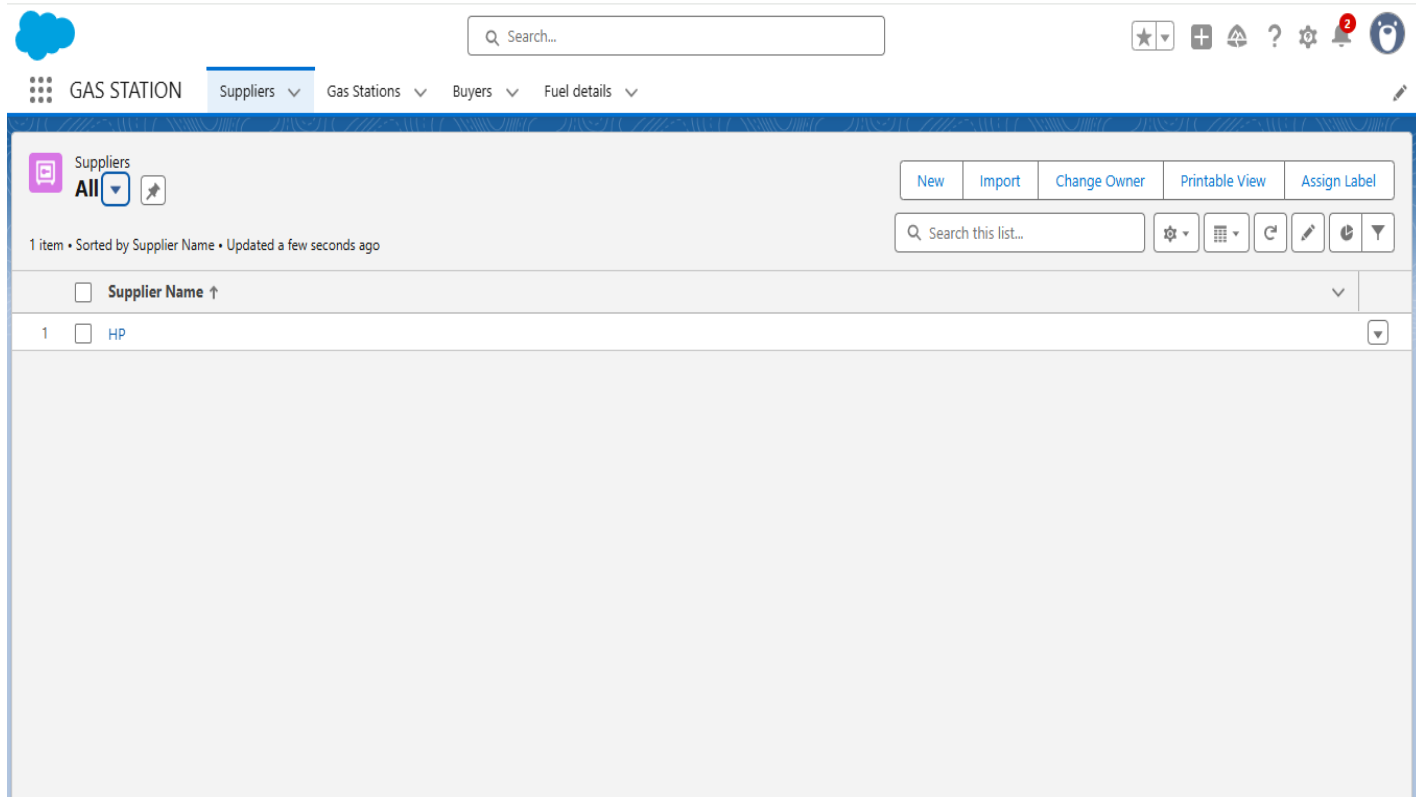
Purpose:

To build a dedicated workspace in Salesforce that groups all related custom objects, tabs, and utilities for the Gas Station CRM. This app streamlines user navigation and centralizes all tools in one interface.

Procedure:

1. In Salesforce, go to Setup → App Manager → New Lightning App.
2. Enter an App Name "Gas Station App"
3. Configure the app's Navigation Style as Standard Navigation for desktop use.
4. Under Select Items, add the custom objects you created earlier:
 - Buyer
 - Fuel Details
 - Gas Station
 - Supplier
5. Assign the app to the profile "System Administrator" to control who can access it.
6. Review your settings and click Finish to create the app.

Test the app by switching to the App Launcher and opening *Gas Station App* to verify all tabs and objects are visible



Step 5 – Creating Fields for the Custom Objects:

Purpose:

To add specific fields to each custom object so that all necessary details for buyers, suppliers, stations, and fuel records are accurately captured and organized.

Procedure:

1. Go to Setup → Object Manager and select the object you want to modify (e.g., Buyer).
2. Click Fields & Relationships → New.
3. Choose the Field Type (e.g., Text, Picklist, Formula, Number) based on the information you need to store, then click Next.
4. Provide a Field Label, set properties such as length or picklist values, and configure help text if needed.
5. Assign field-level security and add the field to the appropriate page layouts.
6. Repeat this process for all fields across the four custom objects.

Fields to be Created:

For Buyer__c:

- First Name (Text)
- Last Name (Text)
- Customer Name (Formula: combines First and Last Name)
- Phone Number (Text, with validation for the correct format)

- Email (Email)
- Vehicle Type (Picklist)
- Fuel Filled in Vehicle (Number)
- Mode of Payment (Picklist)
- Amount Paid (Formula)

Setup > Object Manager

Buyer

Details

Fields & Relationships

13 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Amount Paid	Amount_Paid__c	Formula (Number)		
Buyer Name	Name	Auto Number		✓
Created By	CreatedById	Lookup(User)		
Customer Name	Customer_Name__c	Formula (Text)		
email	email__c	Email		
First name	First_name__c	Text(20)		
Fuel filled in vehicle	Fuel_filled_in_vehicle__c	Number(18, 0)		
Gas Station name	Gas_Station_name__c	Master-Detail(Gas Station)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Last name	Last_name__c	Text(20)		
Mode of payment	Mode_of_payment__c	Picklist		
Phone number	Phone_number__c	Phone		

For Fuel_Details__c:

- Fuel Supplied (Number)
- Supplier Name (Master-Detail)
- Gas Station (Master-Detail)

Setup > Object Manager

Fuel details

Details

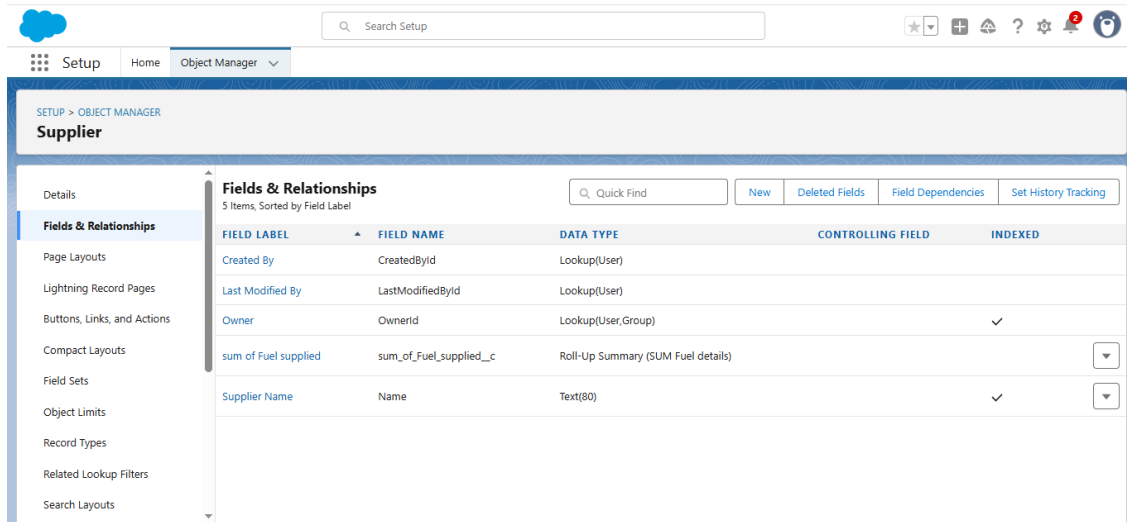
Fields & Relationships

6 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Fuel details Name	Name	Auto Number		✓
Fuel Supplied	Fuel_Supplied__c	Number(5, 0)		
Gas Station	Gas_Station__c	Master-Detail(Gas Station)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Supplier Name	Supplier_Name__c	Master-Detail(Supplier)		✓

For Supplier__c:

- Sum of Fuel Supplied (Roll-Up Summary)

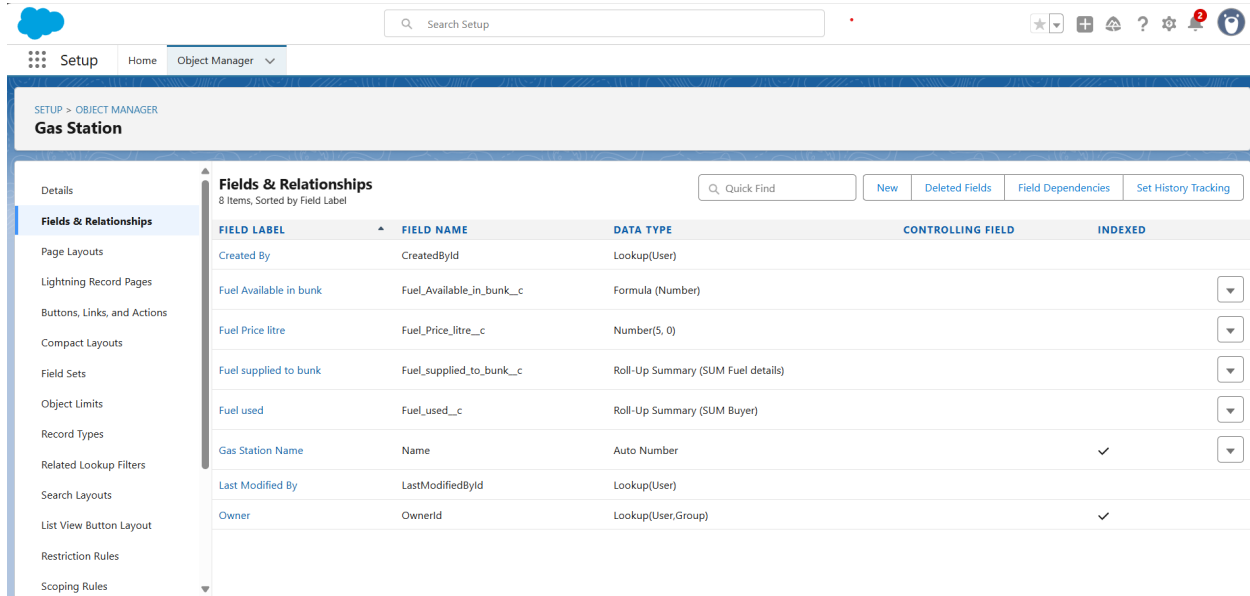


The screenshot shows the Salesforce Setup interface for the 'Supplier' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Search Layouts. The main content area is titled 'Supplier' and shows the 'Fields & Relationships' section. It displays a table with 5 items, sorted by Field Label. The table has columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are: Created By (CreatedById, Lookup(User)), Last Modified By (LastModifiedById, Lookup(User)), Owner (Ownerid, Lookup(User,Group)), sum of Fuel supplied (sum_of_fuel_supplied__c, Roll-Up Summary (SUM Fuel details)), and Supplier Name (Name, Text(80)).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	Ownerid	Lookup(User,Group)		✓
sum of Fuel supplied	sum_of_fuel_supplied__c	Roll-Up Summary (SUM Fuel details)		
Supplier Name	Name	Text(80)		✓

For Gas_Station__c:

- Fuel Price per Liter (Number)
- Fuel Supplied to Bunk (Roll-Up Summary)
- Fuel Used (Roll-Up Summary)
- Fuel Available in Bunk (Formula)



The screenshot shows the Salesforce Setup interface for the 'Gas Station' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Gas Station' and shows the 'Fields & Relationships' section. It displays a table with 8 items, sorted by Field Label. The table has columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are: Created By (CreatedById, Lookup(User)), Fuel Available in bunk (Fuel_Available_in_bunk__c, Formula (Number)), Fuel Price litre (Fuel_Price_litre__c, Number(5, 0)), Fuel supplied to bunk (Fuel_supplied_to_bunk__c, Roll-Up Summary (SUM Fuel details)), Fuel used (Fuel_used__c, Roll-Up Summary (SUM Buyer)), Gas Station Name (Name, Auto Number), Last Modified By (LastModifiedById, Lookup(User)), and Owner (Ownerid, Lookup(User,Group)).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Fuel Available in bunk	Fuel_Available_in_bunk__c	Formula (Number)		
Fuel Price litre	Fuel_Price_litre__c	Number(5, 0)		
Fuel supplied to bunk	Fuel_supplied_to_bunk__c	Roll-Up Summary (SUM Fuel details)		
Fuel used	Fuel_used__c	Roll-Up Summary (SUM Buyer)		
Gas Station Name	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	Ownerid	Lookup(User,Group)		✓

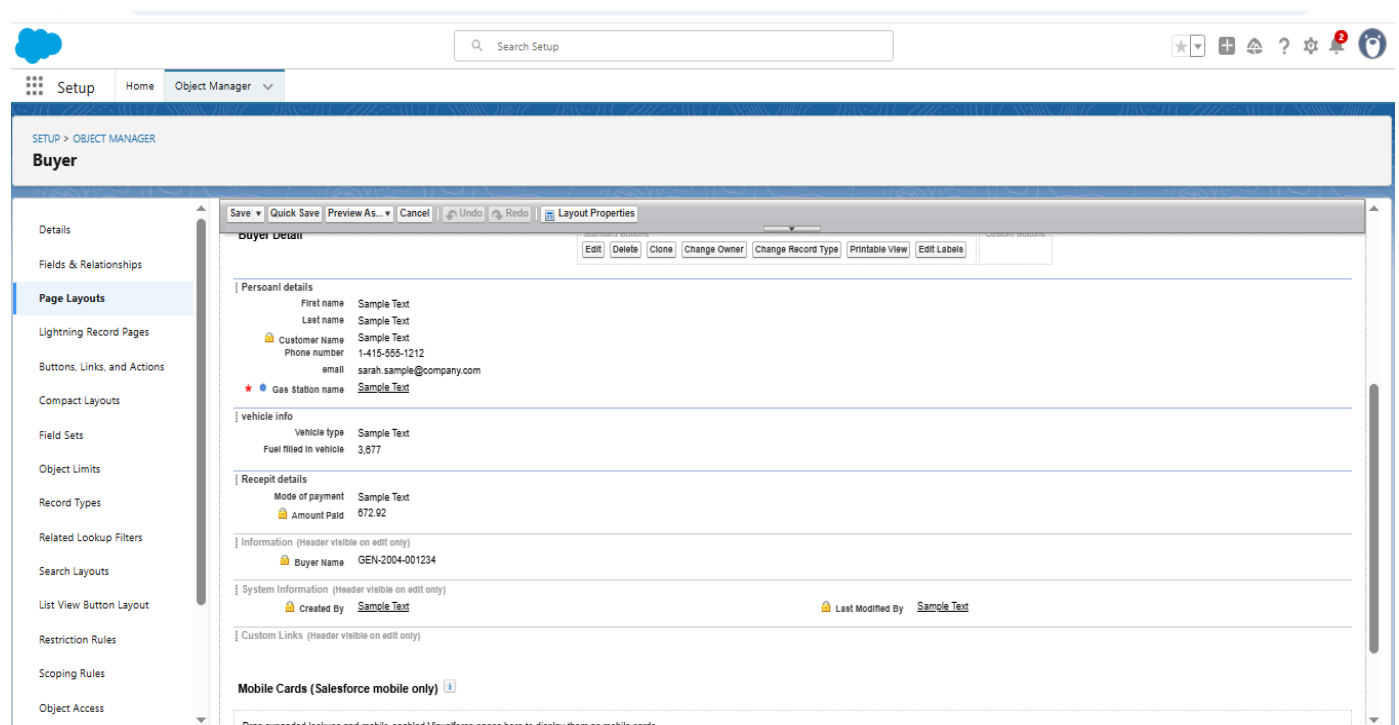
Step 6 – Creating Page Layouts for the Objects:

Purpose:

To organize fields and sections on each record's page so that data entry and viewing are intuitive, structured, and user-friendly.

Procedure:

1. In Salesforce, go to Setup → Object Manager and select the object you wish to modify (e.g., Buyer).
2. Click Page Layouts → New to create a fresh layout.
3. Drag and drop fields into logical sections (e.g., Personal Details, Vehicle Information, Payment Details).
4. Add new sections using the *Section* component to group related fields.
5. Arrange fields to prioritize frequently used information at the top.



Step 7 – Creating Profiles:

Purpose:

To control user permissions and define what actions different roles (e.g., Manager, Sales Executive, Salesperson) can perform within the Gas Station CRM. Profiles ensure security and prevent unauthorized access to sensitive data.

Procedure:

1. In Salesforce, go to Setup → Profiles.
2. Choose an existing profile to clone:
 - Clone Standard User for the Manager profile.
 - Clone Salesforce Platform User for the Sales Executive and Salesperson profiles.
3. Provide a clear Profile Name (e.g., *Manager Profile*, *Sales Executive Profile*, *Salesperson Profile*), then click Save.
4. Open each profile and click Edit to:
 - Configure Custom App Settings so users can access the *Gas Station App*.
 - Set Object Permissions for Buyer, Fuel_Details, Gas_Station, and Supplier objects according to role requirements.
 - Adjust Field-Level Security to restrict sensitive fields if necessary.
 - Review Session Settings and Password Policies to enforce security standards.
5. Assign the profiles to test users or your own account for validation.

The screenshot displays the Salesforce Setup interface for editing a profile. The left sidebar shows the navigation menu with 'Setup', 'Home', and 'Object Manager'. The 'Profiles' section is selected. The main content area shows the 'Profile Edit' page for the 'Manager' profile. The 'Name' field is set to 'Manager' and the 'User License' is 'Salesforce'. The 'Custom Profile' checkbox is checked. Below, the 'Custom App Settings' table is visible, showing various standard Salesforce apps with 'Visible' and 'Default' checkboxes.

App	Visible	Default
All Tabs (standard__AllTabSet)	☒	☐
Analytics Studio (standard__Insights)	☒	☐
App Launcher (standard__AppLauncher)	☒	☐
Approvals (standard__Approvals)	☒	☐
Automation (standard__FlowsApp)	☒	☐
Bot Solutions (standard__LightningBot)	☒	☐
Community (standard__Community)	☒	☐
Contact (standard__Contact)	☒	☐
Data Cloud (standard__Audience360)	☒	☐
My Service Journey (standard__MSJApp)	☒	☐
Queue Management (standard__QueueManagement)	☒	☐
Sales (standard__LightningSales)	☒	☐
Sales (standard__Sales)	☒	☐
Sales Cloud Mobile (standard__SalesCloudMobile)	☒	☐
Sales Console (standard__LightningSalesConsole)	☒	☐
Salesforce Chatter (standard__Chatter)	☒	☐
Salesforce Scheduler Setup (standard__LightningScheduler)	☐	☐
Sample Console (standard__ServiceConsole)	☐	☐

Search Setup

Setup Home Object Manager

profiles

Users

Profiles

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Profiles

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields
Buyers	☒	☐	☐	☐	☐	☐	☐
Fuel details	☒	☐	☐	☐	☐	☐	☐
Gas Stations	☒	☐	☒	☐	☐	☐	☐
Suppliers	☒	☐	☐	☐	☐	☐	☐

Session Settings

Session Times Out After: 2 hours of inactivity

Session Security Level Required at Login: --None--

Password Policies

User passwords expire in: 90 days

Enforce password history: 3 passwords remembered

Minimum password length: 8

Password complexity requirement: Must include alpha and numeric characters

Password question requirement: Cannot contain password

Maximum invalid login attempts: 10

Lockout effective period: 15 minutes

Step 8 – Creating Roles & Hierarchy:

Purpose:

To establish a reporting structure in Salesforce that controls record-level access. Roles define who can view or manage records based on their position within the organization, ensuring secure and organized data sharing.

Procedure

1. In Salesforce, navigate to Setup → Roles → Set Up Roles.
2. Click Expand All to view the default hierarchy.
3. Under the relevant parent node, click Add Role.
4. Enter a Role Label (e.g., *Manager*). The Role Name will auto-populate. Click Save.
5. Create additional roles:
 - **Sales Executive** under Manager.
 - **Salesperson** under Sales Executive.
6. Verify the hierarchy matches the reporting structure:
 - Manager → Sales Executive → Salesperson.
7. Assign the created roles to their corresponding users for proper access control.

Setup Roles

Creating the Role Hierarchy

You can build on the existing role hierarchy shown on this page. To insert a new role, click **Add Role**.

Your Organization's Role Hierarchy Show in tree view

lendi

- Add Role**
- CEO** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - CFO** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - COO** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - Manager** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - sales executive** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - sales person** [Edit](#) [Del](#) [Assign](#)
 - Add Role**
 - SVP_Customer Service & Support** [Edit](#) [Del](#) [Assign](#)
 - Add Role**

Step 9 – Creating Users:

Purpose:

To provide individual login credentials and role assignments for team members, enabling them to access the Gas Station CRM according to their responsibilities.

Procedure:

1. In Salesforce, go to Setup → Users → Users.
2. Click New User to open the user creation form.
3. Fill in the following details:
 - **First Name** and **Last Name**.
 - **Alias** (short identifier).
 - **Email Address** (for login and notifications).
 - **Username** (must be unique across Salesforce orgs, formatted as an email).
 - **Nickname** (auto-populated or custom).
 - **Role** (e.g., Manager, Sales Executive, Salesperson).

- **User License** (Salesforce or Salesforce Platform, depending on profile).
 - **Profile** (assign Manager, Sales Executive, or Salesperson profile created earlier).
4. Set **Active** to true to enable login.
 5. Click **Save**, and Salesforce will email the user with login credentials.
 6. Repeat the process for each team member.

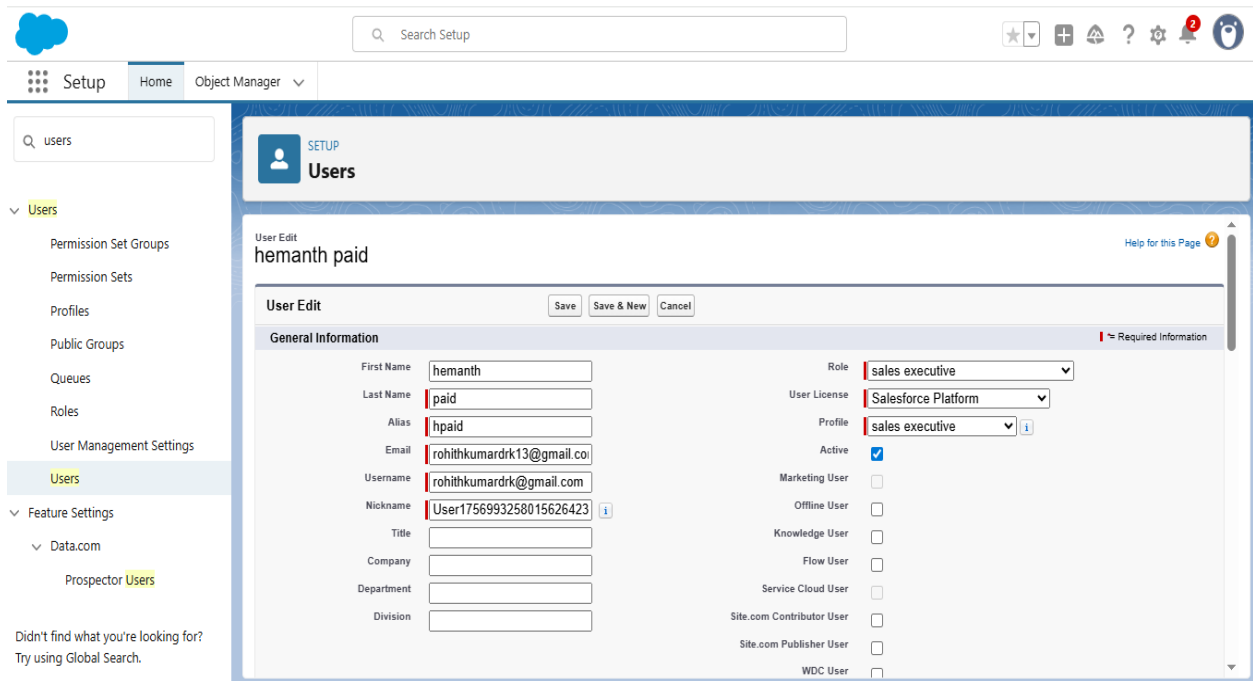
Users:

Manager: Niklaus Mikaelson – Role: Manager Profile.

The screenshot displays the Salesforce Setup interface for editing a user. The top navigation bar includes the Salesforce logo, a search bar, and various utility icons. The left sidebar shows the 'Setup' menu with 'Users' selected. The main content area is titled 'User Edit' and shows the details for 'Niklaus Mikaelson'. The 'General Information' section includes fields for First Name, Last Name, Alias, Email, Username, Nickname, Title, Company, Department, and Division. The 'Role' is set to 'Manager', 'User License' is 'Salesforce', and 'Profile' is 'Manager'. The 'Active' checkbox is checked. Other user types like Marketing User, Offline User, Knowledge User, Flow User, Service Cloud User, Site.com Contributor User, Site.com Publisher User, and WDC User are listed with unchecked checkboxes.

User Edit	
Save Save & New Cancel	
General Information	
First Name	Niklaus
Last Name	Mikaelson
Alias	nmika
Email	22kd1a0229@lendi.edu.in
Username	22kd1a0229@lendi.edu.in
Nickname	User1756992356580432090
Title	
Company	
Department	
Division	
Role	Manager
User License	Salesforce
Profile	Manager
Active	☒
Marketing User	☐
Offline User	☐
Knowledge User	☐
Flow User	☐
Service Cloud User	☐
Site.com Contributor User	☐
Site.com Publisher User	☐
WDC User	☐

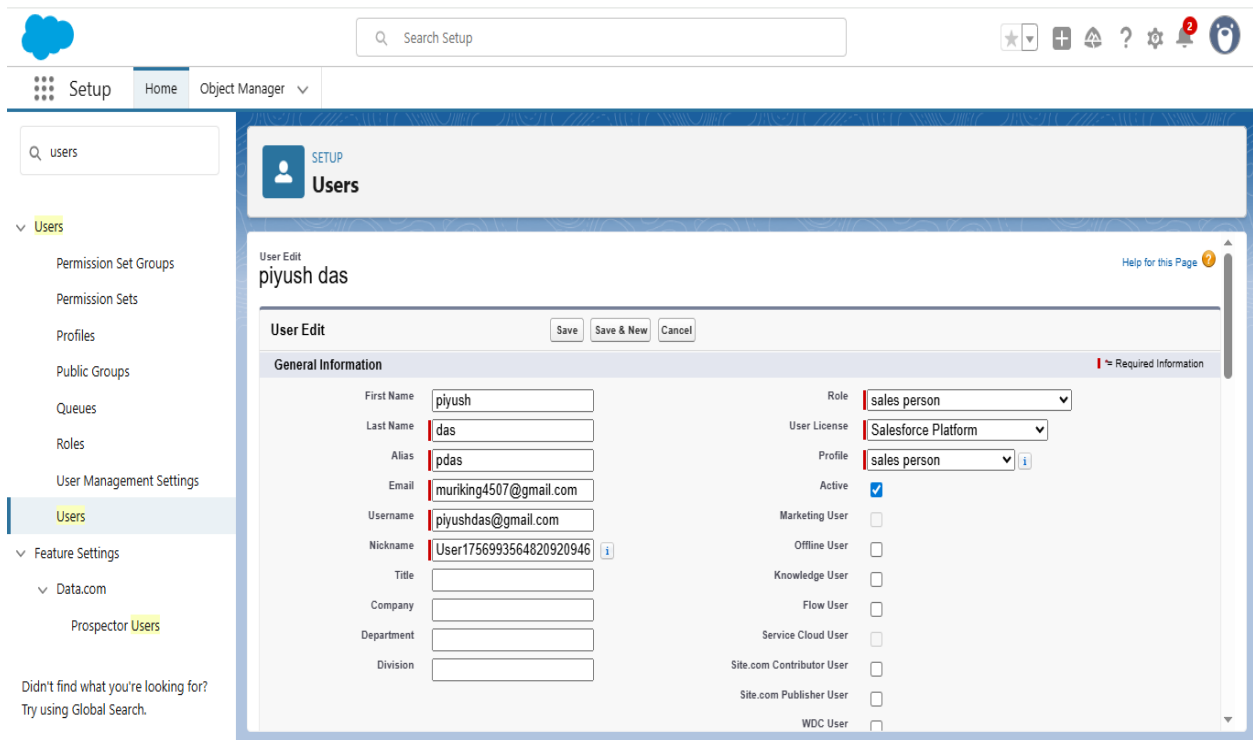
Sales Executive: hemanth paid – Role: Sales Executive Profile.



The screenshot shows the Salesforce Setup interface for editing a user. The left sidebar contains navigation links: Setup, Home, Object Manager, Users, Permission Set Groups, Permission Sets, Profiles, Public Groups, Queues, Roles, and User Management Settings. The main content area is titled 'User Edit hemanth paid' and includes a 'General Information' section. The user's details are as follows:

Field	Value
First Name	hemanth
Last Name	paid
Alias	hpaid
Email	rohithkumardrk13@gmail.co
Username	rohithkumardrk@gmail.com
Nickname	User1756993258015626423
Title	
Company	
Department	
Division	
Role	sales executive
User License	Salesforce Platform
Profile	sales executive
Active	☒
Marketing User	☐
Offline User	☐
Knowledge User	☐
Flow User	☐
Service Cloud User	☐
Site.com Contributor User	☐
Site.com Publisher User	☐
WDC User	☐

Salesperson: Piyush Das – Role: Salesperson Profile



The screenshot shows the Salesforce Setup interface for editing a user. The left sidebar contains navigation links: Setup, Home, Object Manager, Users, Permission Set Groups, Permission Sets, Profiles, Public Groups, Queues, Roles, and User Management Settings. The main content area is titled 'User Edit piyush das' and includes a 'General Information' section. The user's details are as follows:

Field	Value
First Name	piyush
Last Name	das
Alias	pdas
Email	muriking4507@gmail.com
Username	piyushdas@gmail.com
Nickname	User1756993564820920946
Title	
Company	
Department	
Division	
Role	sales person
User License	Salesforce Platform
Profile	sales person
Active	☒
Marketing User	☐
Offline User	☐
Knowledge User	☐
Flow User	☐
Service Cloud User	☐
Site.com Contributor User	☐
Site.com Publisher User	☐
WDC User	☐

Step 10 – Creating Permission Sets:

Purpose:

To grant additional permissions to users without altering their assigned profiles. Permission sets offer flexibility by providing temporary or specialized access to specific objects or actions in the Gas Station CRM.

Procedure:

1. In Salesforce, navigate to Setup → Permission Sets → New.
2. Enter a Label for the permission set "P1" The API name auto-populates.
3. Select the User License that matches the users who will receive this permission set.
Click Save.
4. Under App Permissions or Object Settings, configure the additional access rights needed—for example, granting Sales Executives extra privileges for certain objects.
5. Click Manage Assignments → Add Assignments, select the user Sales Executives and click Assign.

The screenshot displays the Salesforce Setup interface. The left sidebar shows the navigation menu with 'Setup' selected, and 'Permission Sets' highlighted under 'Users'. The main content area shows the 'Current Assignments' for Permission Set 'P1'. The breadcrumb trail is '... > SETUP > PERMISSION SET 'P1''. Below the header, there is a table with columns: Full Name, Active, Role, Profile, User License, and Expires On. The table contains one row for the user 'hemant paid', who is active and assigned the 'sales executive' role and profile, using the 'Salesforce Platform' user license. An 'Add Assignment' button is visible in the top right corner of the table area.

☐	Full Name ↑	Active	Role	Profile	User License	Expires On
☐	hemant paid	✓	sales executive	sales executive	Salesforce Platform	

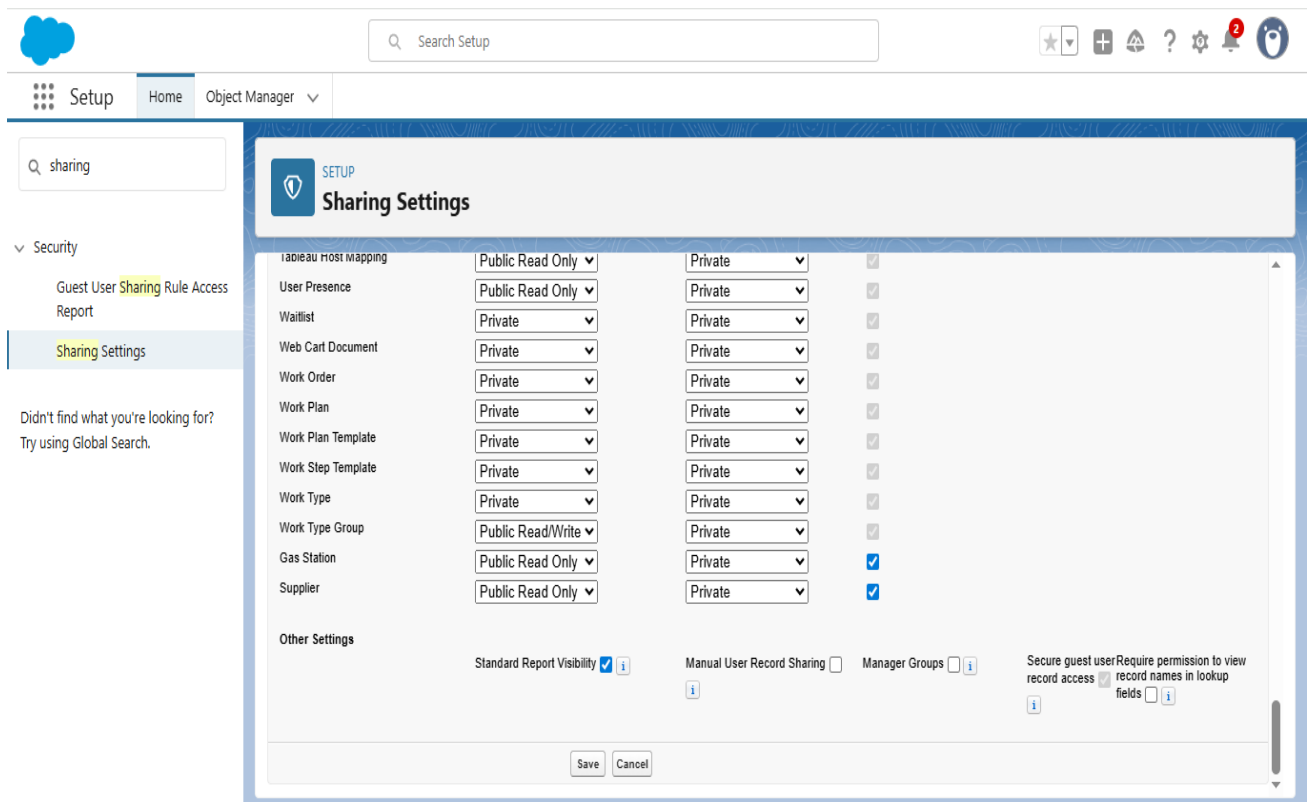
Step 11 – Configuring Organization-Wide Defaults (OWD):

Purpose:

To define the baseline level of data access for all users. OWD settings ensure that records users do not own are protected, while still allowing visibility where appropriate.

Procedure

1. In Salesforce, go to Setup → Sharing Settings.
2. Locate the Organization-Wide Defaults section.
3. For each custom object Gas Station, Supplier sets the default access level:
 - Public Read-Only – Users can view records but cannot edit or delete them.
 - Keep more sensitive objects or data as Private if needed.
4. Click Save to apply your changes.



The screenshot shows the Salesforce Setup interface with the Sharing Settings page open. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'Sharing Settings' and contains a table of sharing settings for various objects. The table has columns for the object name, the default sharing setting (Public Read Only, Private, or Public Read/Write), and a checkbox for 'Secure guest user'. The 'Gas Station' and 'Supplier' objects are highlighted with blue checkmarks in the 'Secure guest user' column.

Object	Default Sharing Setting	Secure guest user
Account	Public Read Only	☒
User Presence	Public Read Only	☒
Waitlist	Private	☒
Web Card Document	Private	☒
Work Order	Private	☒
Work Plan	Private	☒
Work Plan Template	Private	☒
Work Step Template	Private	☒
Work Type	Private	☒
Work Type Group	Public Read/Write	☒
Gas Station	Public Read Only	☒
Supplier	Public Read Only	☒

Other Settings

Standard Report Visibility ☒ [i](#) Manual User Record Sharing ☐ [i](#) Manager Groups ☐ [i](#) Secure guest user Require permission to view record access ☒ [i](#) record names in lookup fields ☐ [i](#)

Save Cancel

Step 12 – Creating, Viewing, and Deleting Records:

Purpose:

To validate that your custom objects, fields, permissions, and layouts function correctly by performing basic record operations in the Gas Station CRM

Procedure:

Creating Records:

1. Open the Gas Station App from the App Launcher.
2. Navigate to a tab (e.g., Buyer, Fuel Details, Gas Station, or Supplier).
3. Click New to create a record.
4. Fill in the required fields, such as Buyer Name, Fuel Details, or Supplier Info.
5. Click Save and confirm the record appears in the list view.

The screenshot displays the Salesforce Lightning interface for the 'GAS STATION' app. A modal window titled 'New Fuel details' is open, showing a form with the following fields:

- Information** (Section Header)
- Fuel details Name** (Text field)
- * Supplier Name** (Text field with a dropdown arrow, containing 'HP')
- * Gas Station** (Text field with a dropdown arrow, containing 'Gas-001')
- Fuel Supplied** (Text field with a dropdown arrow, containing '1000')

At the bottom of the modal, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The background shows the 'Fuel details' list view with a 'Recently Viewed' section and a table containing one item, 'fuel-002'.

Viewing Records:

1. From the same tab, open the list view.
2. Click on a record's Name to view its details.
3. Verify that all sections and fields (e.g., Customer Info, Fuel Transaction Details) display correctly based on the page layouts.

Search...

GAS STATION Suppliers Gas Stations Buyers Fuel details

Fuel details
fuel-003

New Contact Edit New Opportunity

Related **Details**

Fuel details Name
fuel-003

Supplier Name
HP

Gas Station
Gas-001

Fuel Supplied
1,000

Created By
dima rohith kumar, 9/24/2025, 2:54 AM

Last Modified By
dima rohith kumar, 9/24/2025, 2:54 AM

Deleting Records:

1. Select a record you wish to remove.
2. Click the Delete option.
3. Confirm the deletion when prompted.
4. Verify that deletion follows any validation rules or Apex triggers you configured (e.g., preventing deletion under certain conditions).

Search...

GAS STATION Suppliers Gas Stations Buyers Fuel details

Fuel details
fuel-003

New Contact Edit New Opportunity

Related **Details**

Fuel details Name
fuel-003

Supplier Name
HP

Gas Station
Gas-001

Fuel Supplied
1,000

Created By
dima rohith kumar, 9/24/2025, 2:54 AM

Last Modified By
dima rohith kumar, 9/24/2025, 2:54 AM

Delete Fuel details

Are you sure you want to delete this Fuel details?

Cancel Delete

Step 13 – Creating Reports:

Purpose:

To analyze and display Salesforce data in a structured, meaningful way. Reports help track fuel usage, customer activity, and sales performance, enabling informed decision-making for gas station operations.

Procedure:

1. From Salesforce's App Launcher, open the Reports tab.
2. Click New Report.
3. Choose the report type (e.g., *Fuel Details with Buyers* or *Gas Station with Suppliers*).
4. Add the necessary fields and filters—examples include:
 - Buyer Name, Fuel Quantity, and Amount Paid.
 - Supplier Name and Fuel Supplied.
5. Group data as needed (e.g., by Buyer or Supplier) and add summaries or totals.
6. Adjust the report format (Tabular, Summary, Matrix) based on your analysis needs.
7. Click Save & Run, provide a descriptive report name, and select a folder for storage.
8. Test the report by running it to ensure it displays accurate data.

Report: Gas Stations with Buyers
Amount range

Total Records: 10, Total Fuel filled in vehicle: 184, Total Amount Paid: 21,344.00

☐ Fuel Available in tank	☐ Customer Name	☐ Fuel filled in vehicle	☐ Amount Paid
☐ 2,816.00 (10)	akula chiru (1)	45	5,220.00
	Subtotal	45	5,220.00
	AP CHARAN SAI (1)	2	232.00
	Subtotal	2	232.00
	Bhagel Rohith (1)	24	2,784.00
	Subtotal	24	2,784.00
	bobbadi hemanth (1)	10	1,160.00
	Subtotal	10	1,160.00
	boni sumanth (1)	15	1,740.00
	Subtotal	15	1,740.00
	chilli mukesh (1)	15	1,740.00
	Subtotal	15	1,740.00
	dinku anil kumar (1)	55	6,380.00
	Subtotal	55	6,380.00
	khan alhaaf (1)	1	116.00
	Subtotal	1	116.00
	KOYYA SHANMURGH (1)	5	580.00
	Subtotal	5	580.00
	vidya sagar (1)	12	1,392.00
	Subtotal	12	1,392.00
	Subtotal	184	21,344.00
	Total (10)	184	21,344.00

Row Counts: ☒ Detail Rows ☒ Subtotals ☒ Grand Total

Step 14 – Creating Dashboards:

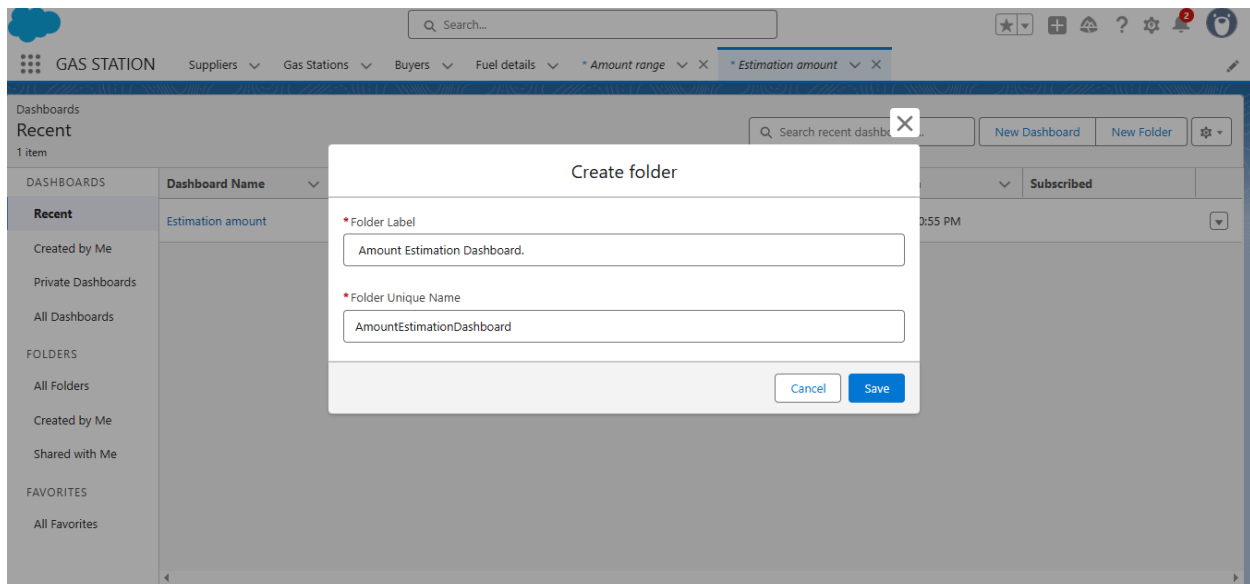
Purpose:

To represent key performance indicators visually and report data in an interactive format. Dashboards allow managers and staff to assess fuel usage rapidly, sales trends, and customer activity for better decision-making.

Procedure:

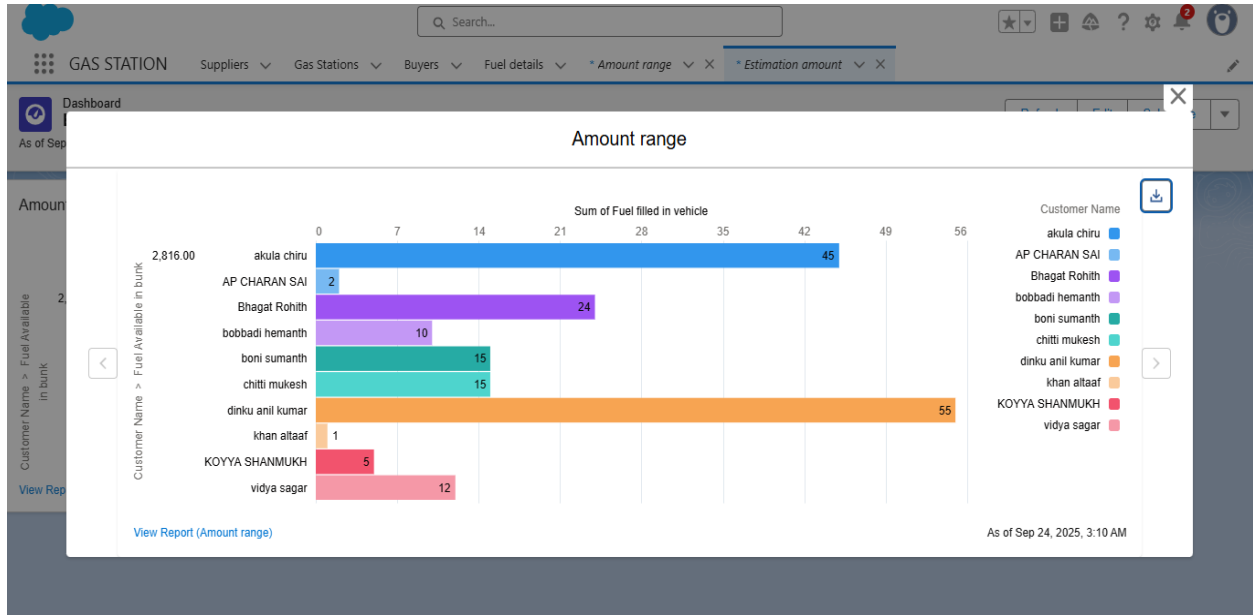
Create Dashboard Folder:

1. Click on the App Launcher and search for Dashboard.
2. Click on the Dashboard tab.
3. Click New Folder and enter the folder label as Amount Estimation Dashboard.
4. The Folder Unique Name will auto-populate automatically.
5. Click Save.



Create Dashboard:

1. In Salesforce, open the Dashboards tab from the App Launcher.
2. Click New Dashboard.
3. Enter a Dashboard Name Estimation amount and choose a folder for storage.
4. Click Create, then + Component to add visual elements.
5. Select the report you created earlier (e.g., Fuel Usage Report, Supplier Performance Report).
6. Choose a visualization type (e.g., bar chart, pie chart, line graph) that best represents the data.
7. Click save and done.



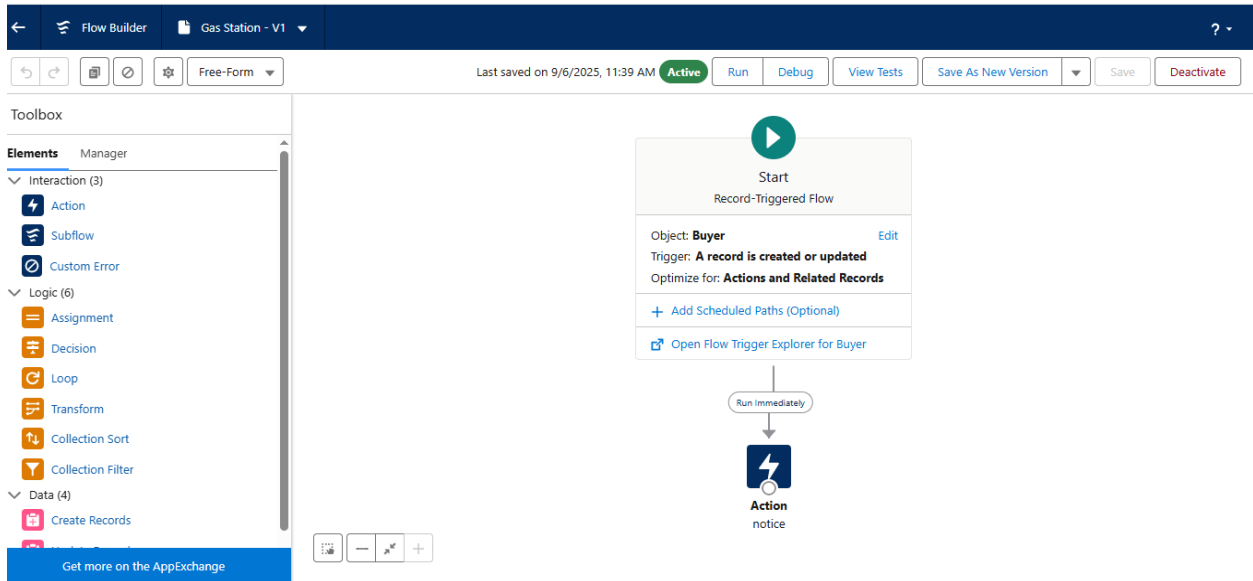
Step 15 – Creating Flows:

Purpose:

To automate key business processes and reduce manual work. In this project, a record-triggered flow is used to automatically send an email receipt to customers whenever a buyer record is created or updated.

Procedure:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow, choose Buyer as the object, set the trigger to A record is Created or Updated, optimize for Actions and Related Records, and click Done.
3. Switch to Free-Form layout.
4. In Manager, create a Text Template resource (`email body`) and enter the email content with merge fields for Customer Name, Amount Paid, Vehicle Type, and Fuel Intake. Set view to Plain Text.
5. Drag an Action element into the canvas, search for Send Email, and select it.
6. Label it `Notice`, select the `email body` template, set the recipient email and subject (`Welcome to Gas Station`), then connect it to the Start element.
7. Save the flow, provide a label, and Activate.



Step 16 – Creating Apex Triggers:

Purpose:

To enforce custom business logic that cannot be achieved through standard Salesforce configuration or flows.

Procedure:

Apex triggers are custom code that execute before or after events on records.

Handler (FuelRecordHandler.apxc)

- beforeDeleteInfo → Stops deletion of Fuel Details if supplier qty > 500.
- beforeDeleteGas → Ensures Gas Station fuel price > 50 before insert.

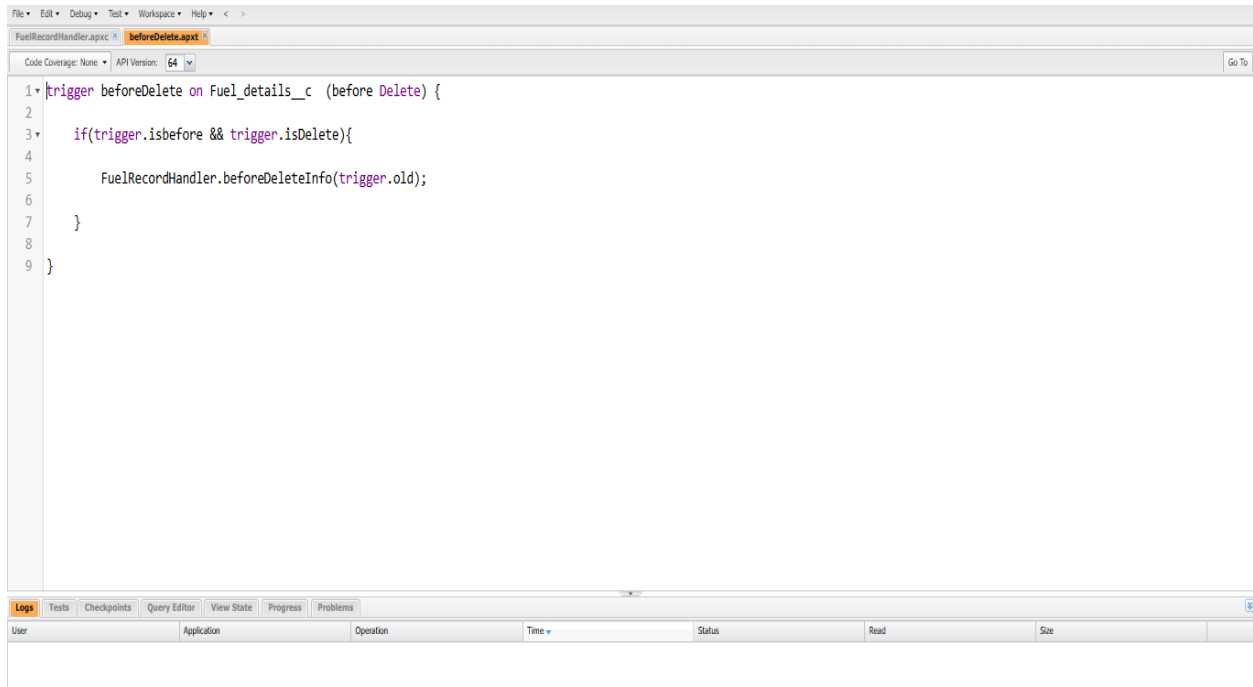
```

1 public class FuelRecordHandler {
2
3     public static void beforeDeleteInfo(list<Fuel_Details__c> fuelList){
4
5         //fuelList = [select Id from Fuel_Details__c];
6
7         for(Fuel_Details__c ful : fuelList){
8
9             if(ful.Fuel_Supplied__c > 500){
10
11                 ful.addError('you cannot delete the fuel details record because it is associated with supplier and Gas station records');
12
13             }
14
15         }
16     }
17
18     public static void beforeDeleteGas(list<Gas_Station__c> gasList){
19
20         //fuelList = [select Id from Fuel_Details__c];
21
22         for(Gas_Station__c gas : gasList){
23
24             if(gas.Fuel_Price_litre__c <= 50){
25
26                 gas.addError('enter the fuel price before saving the record, Minimum price should be 50');
27
28             }
29
30         }
31     }
32
33 }
34
35

```

Trigger: Fuel Details (before delete)

Prevents deleting Fuel Details if linked records exist



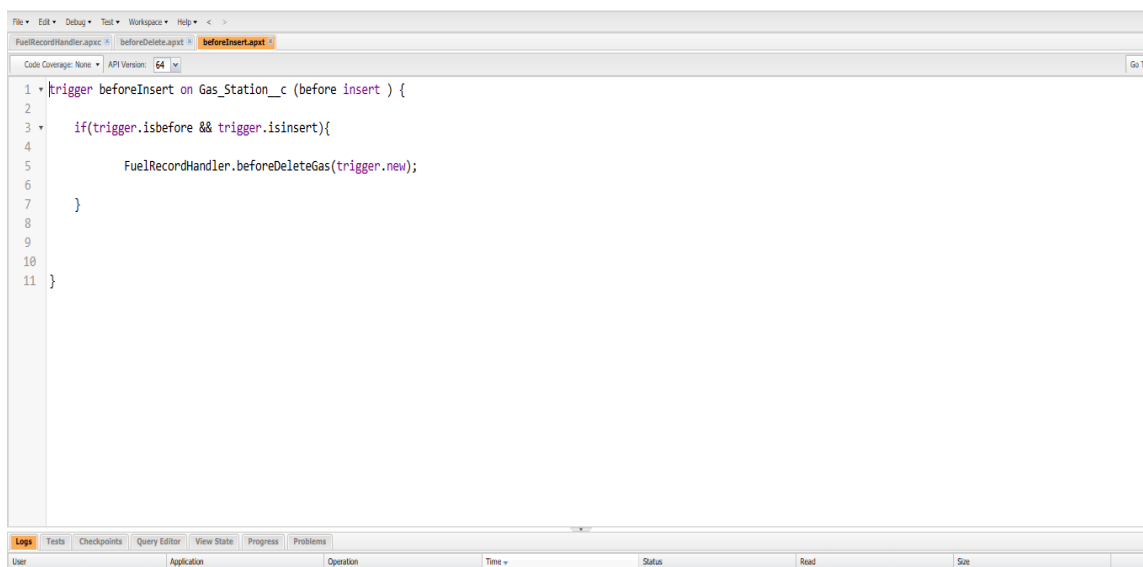
The screenshot shows an IDE window with the file 'FuelRecordHandler.apex' open. The editor displays the following Apex code for a beforeDelete trigger:

```
1 trigger beforeDelete on Fuel_details__c (before Delete) {  
2  
3     if(trigger.isbefore && trigger.isDelete){  
4  
5         FuelRecordHandler.beforeDeleteInfo(trigger.old);  
6  
7     }  
8  
9 }
```

The IDE interface includes a menu bar (File, Edit, Debug, Test, Workspace, Help), a toolbar with 'Code Coverage: None' and 'API Version: 64', and a 'Go To' button. The bottom status bar shows tabs for 'Logs', 'Tests', 'Checkpoints', 'Query Editor', 'View State', 'Progress', and 'Problems'. Below the editor is a table with columns: User, Application, Operation, Time, Status, Read, Size.

Trigger: Gas Station (before insert)

Validates fuel price before record creation.



The screenshot shows an IDE window with the file 'FuelRecordHandler.apex' open. The editor displays the following Apex code for a beforeInsert trigger:

```
1 trigger beforeInsert on Gas_Station__c (before insert ) {  
2  
3     if(trigger.isbefore && trigger.isinsert){  
4  
5         FuelRecordHandler.beforeDeleteGas(trigger.new);  
6  
7     }  
8  
9  
10  
11 }
```

The IDE interface is identical to the previous screenshot, showing the same menu bar, toolbar, and status bar.

Feature Enhancements:

To improve efficiency further and customer satisfaction, the Gas Station CRM can be expanded using Salesforce Einstein capabilities:

- **AI Chatbots (Einstein Bots):** Automate responses to customer inquiries, display current fuel pricing, and issue digital receipts without staff intervention.
- **Personalized Recommendations (Next Best Action):** Suggest discounts or promotions tailored to individual fueling patterns.
- **Automated Communication Logging (Einstein Activity Capture):** Seamlessly record customer emails and interactions for improved tracking.
- **Advanced Visual Analytics (Einstein Analytics):** Deliver detailed performance insights, supplier metrics, and demand forecasts through dynamic dashboards.
- **Customer Behavior Predictions (Prediction Builder):** Forecast customer return frequency and preferences to guide marketing and loyalty initiatives.

Conclusion:

The Gas Station CRM project demonstrates how Salesforce can be tailored to address the specific operational needs of a gas filling station. By implementing custom objects, page layouts, roles, profiles, and permission sets, the system organizes critical data while maintaining secure access for different user roles. Automated processes, such as record-triggered flows and Apex triggers, minimize manual effort and ensure data accuracy.

Reports and dashboards provide clear insights into sales performance, fuel usage, and supplier activity, enabling informed decision-making. Together, these features streamline daily operations, improve collaboration among staff, and enhance the overall customer experience. The project also establishes a scalable foundation for future upgrades, such as AI-driven analytics or chatbot integrations, ensuring the solution can evolve with business growth.